Functional Annotation Report: dapD (Q88MP1, PP_1530) — Tetrahydrodipicolinate N-succinyltransferase, *Pseudomonas putida* KT2440

0. Identity verification

The requested target is UniProt **Q88MP1**, gene **dapD** (ordered locus **PP_1530**) of *Pseudomonas putida* KT2440, annotated as **2,3,4,5-tetrahydropyridine-2,6-dicarboxylate N-succinyltransferase** (tetrahydrodipicolinate N-succinyltransferase; THDP succinyltransferase), **EC 2.3.1.117**. The gene symbol, EC number, protein family ("type 2 tetrahydrodipicolinate N-succinyltransferase"), and InterPro domains (DapD_gammaproteobac. IPR019876; Hexapep IPR001451; THDPS_M IPR032784/IPR038361; Trimer_LpxA-like_sf IPR011004) are **fully self-consistent** and match the biochemically and structurally characterized DapD enzymes described below. The symbol is **not ambiguous**: DapD is a conserved, well-defined bacterial biosynthetic enzyme.

Evidentiary caveat. No study has, to my knowledge, biochemically characterized *P. putida* KT2440 DapD directly. Its function is assigned by (i) automated HAMAP rule MF_02122 and (ii) very strong homology to characterized orthologs. The **closest characterized relative is *Pseudomonas aeruginosa* DapD** (same genus; P 22359568), supplemented by structural/kinetic work on *E. coli*, *Mycobacterium tuberculosis*, *Corynebacterium glutamicum*, and *Serratia marcescens* DapD. All belong to the same family, and the gammaproteobacterial subtype (IPR019876) is shared with *E. coli*, *Serratia*, and *Pseudomonas*. Inferences below are therefore high-confidence homology transfers, explicitly flagged where relevant.

1. Summary

DapD is a **cytoplasmic homotrimeric acyltransferase of the hexapeptide-repeat (left-handed β-helix / LpxA-like) superfamily** that catalyzes the **first committed step of the succinylase branch of the meso-diaminopimelate (meso-DAP)/L-lysine biosynthetic pathway**. It transfers a succinyl group from **succinyl-CoA** onto the amino group of the acceptor derived from **tetrahydrodipicolinate (THDP)** — the ring-opened form L-2-amino-6-oxopimelate — producing **N-succinyl-L-2-amino-6-oxopimelate** + **CoA** (EC 2.3.1.117). This step channels metabolic flux toward meso-DAP, an essential cross-linking residue of Gram-negative peptidoglycan and the immediate precursor of L-lysine.

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2. Primary molecular function: the reaction catalyzed

DapD catalyzes the succinyl-CoA–dependent N-succinylation of the open-chain amino-keto acid generated from THDP:

> **THDP (→ L-2-amino-6-oxopimelate) + succinyl-CoA → N-succinyl-L-2-amino-6-oxopimelate + CoA**

- Beaman et al. (2002) define the reaction precisely: DapD "catalyzes the **succinyl-CoA-dependent acylation of L-2-amino-6-oxopimelate to 2-N-succinyl-6-oxopimelate** as part of the **succinylase branch** of the meso-diaminopimelate/lysine biosynthetic pathway of bacteria, blue-green algae, and plants" (P 11910040).
- A 2025 kinetic/spectroscopic study restates the substrates/products: "DapD catalyzes the reaction of **tetrahydrodipicolinate (THDP) and succinyl-CoA** to form **(S)-2-(3-carboxypropanamido)-6-oxoheptanedioic acid and coenzyme A**" (Hayes et al. 2025, P 40930446).

Chemically, cyclic THDP is in equilibrium with its open-chain amino-keto acid form (2-amino-6-oxopimelate/2-amino-6-oxoheptanedioate); it is the free primary amine of this open form that is succinylated. The succinyl "handle" installed here is retained through the next two steps and later removed, serving as a protecting/activating group that permits regioselective transamination.

Substrate specificity

- **Acyl donor:** succinyl-CoA (not acetyl-CoA). The discrimination between succinyl- and acetyl-CoA — i.e., between the **succinylase** and **acetylase** branches of DAP biosynthesis — is set by "**at least three amino acids that interact with or give accessible active site volume to the cofactor succinyl group**" (P 11910040). DapD's assignment to the succinylase branch (and IPR032784 "THDPS_M") is thus mechanistically grounded.
- **Amino acceptor — stereospecificity:** In the closest characterized ortholog, *P. aeruginosa* DapD "**is specific for the L-stereoisomer of the amino substrate L-2-aminopimelate, and its D-enantiomer acts as a weak inhibitor**"; both enantiomers bind the same pocket, but the D-amino group is **misaligned for nucleophilic attack**, explaining specificity and inhibition (P 22359568). (L-2-aminopimelate is used experimentally as a stable, non-cyclizing analog of the true acceptor.)

Catalytic / kinetic mechanism

- **Ordered ternary-complex ("direct attack") mechanism, not ping-pong.** Crystallographic ternary complexes place the succinyl thioester carbonyl carbon "**in close proximity to the nucleophilic 2-amino group of the acceptor, in support of a direct attack ternary complex mechanism**" (P 11910040) — both substrates bind before chemistry; no covalent acyl-enzyme intermediate.
- **Kinetics:** For *S. marcescens* DapD, initial-velocity patterns are "**consistent with a rapid equilibrium ordered bi bi kinetic mechanism**," and "conversion of the central enzyme

complexes (chemistry) is thought to be the **rate-limiting step**"
(P 40930446).

- **Cooperativity:** In *E. coli* DapD, "the **C-terminal helix undergoes a large rearrangement upon substrate binding, which contributes to cooperativity** in substrate binding" (P 18242192).

2b. Sequence-based validation of the homology assignment (this work)

To quantify the confidence of transferring function from characterized orthologs, I retrieved the UniProt sequences and computed pairwise alignments (BLOSUM62; global Needleman–Wunsch and local Smith–Waterman) between *P. putida* DapD (Q88MP1, 344 aa) and reviewed DapD orthologs:

Ortholog (UniProt)	Length	% identity to Q88MP1
P. aeruginosa PAO1 DapD (G3XD76)	344 aa	**88.4 %** (304/344)
C. glutamicum DapD (Q8NRE3)	297 aa	48.6 %
M. tuberculosis DapD (P9WP21)	317 aa	47.7 %
E. coli K12 DapD (P0A9D8)	274 aa	~21 % global; ~28 % local core

Key conclusions: - *P. putida* DapD is **88.4 % identical** to *P. aeruginosa* DapD — the same-genus enzyme whose crystal structures, L-stereospecificity, and succinyl-CoA usage were experimentally established (P 22359568). This near-identity makes transfer of catalytic function, substrate specificity, and quaternary structure to Q88MP1 **essentially definitive**. - Q88MP1 and G3XD76 have **identical length (344 aa)** and both carry the ~70-residue N-terminal domain characteristic of the **gammaproteobacterial "type 2" DapD subtype** (InterPro IPR019876), consistent with the UniProt "type 2 tetrahydrodipicolinate N-succinyltransferase" family assignment. - The low global identity to *E. coli* DapD is an **architectural difference, not lack of orthology**: the *E. coli* enzyme is a shorter 274-aa form lacking the N-terminal extension; the conserved LβH catalytic core still aligns (~28 % local identity). *M. tuberculosis* and *C. glutamicum* DapD (~48 %) are clear, more distant orthologs.

(Alignments computed in this investigation from UniProt REST-retrieved sequences.)

3. Structure and molecular architecture

DapD is a **homotrimer** built on the **hexapeptide-repeat acyltransferase fold**, exactly matching Q88MP1's InterPro annotations (Hexapep IPR001451; Trimer_LpxA-like_sf IPR011004).

- Each subunit has three domains: "**an N-terminal α-helical domain, a left-handed β-helix (LβH) domain, and a β C-terminal domain**" (C. glutamicum, P 26602189). The LβH is "**the common structural motif of enzymes belonging to the hexapeptide repeat superfamily**" (M. tuberculosis, P 19394346).
- Three active sites form at the **subunit interfaces** of the trimer; succinyl-CoA binds there, and the mobile C-terminal domain closes over the acceptor site (P 26602189 ; P 18242192).
- The trimer can be stabilized by metal ions on the molecular 3-fold axis: in Mtb-DapD, "**the trimeric quaternary structure is stabilized by Mg^{2+} and Na^{+} located on the 3-fold axis**," and the N-terminal domain contains "**an interior water-filled channel**" leading to that axial Mg^{2+} (P 19394346).
- Multiple co-crystal structures (with succinyl-CoA, succinamide-CoA cofactor analog, L- and D-2-aminopimelate, 2-aminopimelate) across orthologs map the cofactor and acceptor subsites and were validated by site-directed mutagenesis of catalytic/binding residues (P 26602189 ;

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P 11910040

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P 22359568

The consistency of *P. putida* DapD's domain content with this fold provides strong structural support for the functional assignment.

Sequence-level confirmation of the LβH fold (this work). Inspection of the Q88MP1 sequence shows the diagnostic **degenerate tandem hexapeptide repeats** of a left-handed β-helix across roughly residues ~200–300 (e.g., …**VSAGV-FVGKGS-DLGGGC-STMGTL-SGGGNI-VIKVGE-GCLIGA-NAGIGI**…), each imperfect [LIV]-G-x-x-x-x coil adding one rung to the β-helix. This independently corroborates the InterPro Hexapep (IPR001451) / Trimer_LpxA-like (IPR011004) assignment and the trimeric acyltransferase architecture.

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4. Pathway context and biological process

DapD initiates the **succinylase (acyl-DAP) branch** — the route used by most Gram-negative bacteria including *Pseudomonas* — of the meso-DAP/L-lysine pathway:

1. Aspartate → … → **L-2,3,4,5-tetrahydrodipicolinate (THDP)** (via DapA/DapB from the common upstream pathway)
2. **THDP + succinyl-CoA → N-succinyl-2-amino-6-oxopimelate + CoA — *DapD (this enzyme)***
3. N-succinyl-aminotransferase (DapC) → N-succinyl-L,L-DAP
4. N-succinyl-L,L-DAP desuccinylase (DapE) → L,L-DAP (+ succinate)
5. DAP epimerase (DapF) → **meso-DAP**
6. **meso-DAP** → incorporated into peptidoglycan **and/or** decarboxylated by DAP decarboxylase (LysA) → **L-lysine**

Organism-specific pathway complement (this work). A UniProt survey of *P. putida* KT2440 (taxid 160488) confirms the **complete succinylase branch is encoded**: dapA (PP_2639, PP_1237; EC 4.3.3.7) → dapB (PP_4725; EC 1.17.1.8) → **dapD (PP_1530; EC 2.3.1.117 — target)** → dapC N-succinyl-LL-DAP aminotransferase (PP_1588; EC 2.6.1.17) → dapE succinyl-DAP desuccinylase (PP_1525; EC 3.5.1.18) → dapF DAP epimerase (PP_5228, PP_3790; EC 5.1.1.7) → lysA DAP decarboxylase (PP_5227, PP_2077; EC 4.1.1.20). Crucially, **no acetylase-branch gene (dapH/dapX) and no meso-DAP dehydrogenase (ddh, EC 1.4.1.16) were detected** — *P. putida* has neither the acetylase alternative nor the single-step dehydrogenase bypass. DapD is therefore the **sole, obligatory committed entry** from THDP into meso-DAP/lysine synthesis and is functionally non-redundant (and very likely essential). Notably, **dapD (PP_1530) lies immediately adjacent to dapE (PP_1525)** in the genome, clustering the succinylating and desuccinylating enzymes and suggesting possible co-regulation.

Physiological outputs. The pathway "**provides lysine for protein synthesis and both lysine and meso-diaminopimelate for construction of the bacterial peptidoglycan cell wall**" (Hutton et al. 2007, P 17579770; see also P 12570844; P 8919551; review P 23504110). In Gram-negative bacteria such as *P. putida*, meso-DAP is the residue that cross-links adjacent peptidoglycan strands, so DapD activity is coupled to **cell-wall integrity, growth, and division**.

Regulation. In *C. glutamicum*, *dapD* is a direct target of the nitrogen master regulator **AmtR**, "**the dapD gene, encoding succinylase involved in m-diaminopimelate synthesis**," linking nitrogen status to lysine biosynthesis (P 19041910). This specific regulatory link is organism-dependent and is not established for *P. putida*, but it illustrates that *dapD* expression can be integrated into nitrogen/amino-acid regulatory networks.

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5. Subcellular localization

DapD is a **soluble cytoplasmic enzyme**. It acts on small, soluble metabolic intermediates and cofactors (THDP-derived amino acid, succinyl-CoA, CoA); it carries no signal peptide, membrane-spanning segment, or lipidation motif, and all characterized orthologs were purified as soluble cytosolic proteins for crystallography and kinetics (e.g., P 18242192; P 19394346; P 26602189). Its function — an early anabolic step of amino-acid/cell-wall-precursor biosynthesis — is carried out in the **cytoplasm**, where the resulting meso-DAP is later handed off to the membrane-associated peptidoglycan-assembly machinery.

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6. Physiological / therapeutic significance

Because the meso-DAP/lysine pathway is **present in bacteria (and plants) but absent in humans**, and because it feeds both protein synthesis and the essential cell wall, its enzymes are attractive antibacterial targets. DapD specifically has been highlighted: "**Since lysine is an essential amino acid to humans, DapD is a potentially safe target for antibiotic therapies**" (P 40930446); the broader strategy is reviewed in P 17579770 P 12570844 P 8919551, and 23504110. The recent finding that DapD can be inactivated by Cu^{2+} (P 40930446) is also relevant to inhibitor/metal-sensitivity considerations. (Note: the essentiality of the succinylase branch varies among species because some bacteria possess alternative branches — dehydrogenase or acetylase; *Pseudomonas* uses the succinylase route.)

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7. Supported vs. refuted hypotheses

Supported (high confidence): - H1 — dapD/Q88MP1 encodes tetrahydrodipicolinate N-succinyltransferase (EC 2.3.1.117). ✔ (family/EC/domain concordance + ortholog biochemistry) - H2 — Reaction: THDP-derived amino acid + succinyl-CoA → N-succinyl product + CoA. ✔ (

P 11910040

P 40930446

) - H3 — Homotrimeric LβH/hexapeptide-repeat (LpxA-like) fold. ✔ (

P 19394346

P 26602189

P 18242192

; matches IPR011004/IPR001451) - H4 — L-stereospecific amino acceptor; succinyl-CoA-specific donor. ✔ (

P 22359568

P 11910040

) - H5 — Ordered ternary-complex mechanism, cytoplasmic localization, feeds meso-DAP/peptidoglycan + lysine. ✔ (

P 11910040

P 40930446

P 17579770

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Refuted / disfavored: - A ping-pong (covalent acyl-enzyme) mechanism — disfavored by ternary-complex structures and ordered bi-bi kinetics (

P 11910040

P 40930446

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- Acetyl-CoA as the physiological donor — DapD is committed to the succinylase branch; donor selectivity resides in a few active-site residues (P 11910040).

8. Limitations and future directions

- **No direct experimental data on *P. putida* KT2440 DapD**; all mechanistic/structural detail is transferred from orthologs. However, this transfer is exceptionally well-supported: the closest characterized relative, *P. aeruginosa* DapD, is **88 % identical** (identical length, same "type 2" architecture), so the residual uncertainty is small. Direct enzymology (kinetic constants, an experimental *P. putida* structure) would provide final confirmation.
- The downstream succinylase-branch genes (dapC PP_1588, dapE PP_1525, dapF, lysA) are present and annotated in *P. putida* KT2440 (confirmed here), but *dapD* essentiality and regulation in *P. putida* have not been experimentally tested; targeted knockout, high-density transposon fitness (RB-TnSeq), and expression studies would establish them directly.
- Precise active-site residues in Q88MP1 could be mapped by sequence/structure alignment to the mutagenesis-validated positions in *E. coli*/*C. glutamicum*/*P. aeruginosa* DapD.

Key references

- P 11910040 — Beaman et al., acyl-group specificity & direct-attack ternary-complex mechanism.
- P 22359568 — Schnell et al., *P. aeruginosa* DapD structure & L-stereospecificity (closest relative).
- P 19394346 — Schuldt et al., *M. tuberculosis* DapD structure, trimer, LβH fold, axial metals.
- P 18242192 — Nguyen et al., *E. coli* DapD structure, C-terminal helix rearrangement & cooperativity.

- 26602189 — Sagong & Kim, *C. glutamicum* DapD structure with succinyl-CoA and substrate analog.
- 40930446 — Hayes et al. 2025, *S. marcescens* DapD kinetics (ordered bi-bi) and Cu inactivation.
- 19041910 — Buchinger et al., *dapD* as an AmtR (nitrogen) regulatory target.
- 17579770 / 12570844 / 8919551 / 23504110 — reviews: DAP/lysine pathway as antibacterial target.